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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 07/15/2018 |
| **PI Name:** | Lawler, John Ph.D. | | |
| **Project Title:** | Redox Regulation of nNOS Translocation and Muscle Atrophy During Mechanical Unloading | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
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| **Zip Code:** | 77843 | **Congressional District:** | 17 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2011 Space Biology NNH11ZTT002N |
| **Start Date:** | 08/24/2012 | **End Date:** | 11/15/2018 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Extended to 11/15/2018 (latest end date was 8/22/2018 before this extension) per F. Hernandez/ARC (Ed., 10/16/18)<br>NOTE: Extended to 8/22/2018 (previously extended in 2016 to 8/22/2017) perNSSC information (Ed., 10/6/17)<br>NOTE: Extended to 8/22/2017 per F. Hernandez/ARC and NSSC information (Ed., 8/30/16)<br>NOTE: Extended to 8/23/2016 per A. Chu/ARC and NSSC information (Ed., 6/16/15)<br>NOTE: Grant number changed to NNX13AE45G, per A. Chu via NSSC (Ed., 1/29/13)<br>NOTE: Grant number changed to NNX12AR62G, per A. Chu via NSSC (Ed., 9/28/12)<br>NOTE: Period of performance (previously 6/27/12-6/26/15) and grant number (previously NNX12AL23G) changed per A. Chu/ARC and NSSC information (Ed., 9/12/2012) | | |
| **Key Personnel Changes/Previous PI:** | 2018: Jeff Hord received his PhD. He is currently an NIH Research Fellow with Kevin Campbell's laboratory at the University of Iowa. | | |
| **COI Name (Institution):** | Fluckey, James Ph.D. ( Texas A&M University ) | | |
| **Grant/Contract No.:** | NNX13AE45G ; NNX12AR62G (previous grant #) | | |

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| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | A critical limitation to astronaut performance during spaceflight missions including those to the International Space Station (ISS) and Mars is the impaired contractile ability of skeletal muscles. Skeletal muscle is a highly specialized tissue that rapidly adapts to dynamic changes in mechanical loading by modifying its ultrastructure and mass. Detection and transmittance of loading are vital to muscle integrity and protein turnover, with a nexus associated with the subsarcolemmal cytoskeleton that includes the dystrophin-glycoprotein complex (DGC) and caveolae. This interface between myocytes and the extracellular matrix confers mechanical integrity and activates cell signaling pathways that regulate protein turnover, mitochondrial, muscle fiber cross-sectional area, and fiber phenotype. Mechanical unloading experienced in the microgravity of spaceflight elicits large decrements in force generating capacity via reduction in muscle fiber cross sectional area or atrophy. While unloading-induced atrophy is a function of reduced protein synthesis coupled with increased protein degradation, activation of the transcription factors FoxO3a and nuclear factor-kappaB have drawn significant attention as mediators. Although the upstream triggers are not fully understood, translocation of neuronal nitric oxide synthase (nNOS) from the DGC and sarcolemma to the sarcoplasm is a novel signaling event that stimulates proteolysis, and thus atrophy. New clues can be gleaned from Duchenne and many of the limb-girdle muscular dystrophies, pathologies involving mutations to DGC, and other sarcolemmal proteins. Because MDs and microgravity both increase susceptibility of muscles to damage, oxidative stress, and pro-inflammatory signaling, a shared root cause is suspected: disruption of the subsarcolemmal cytoskeleton. New studies and Preliminary Data indicate the importance of NAD(P)H oxidase (NOX), a sarcolemmal source of oxidative stress, and caveolin-3 in stimulating myopathy in muscular dystrophy models. NOX may be activated upstream by angiotensin II and c-abl, a tyrosine kinase. Further, our pilot data show that unloading-induced alterations in caveolin-3 and nNOS are redox dependent and linked to atrophy and shift in muscle fiber-type. Our central hypothesis is that oxidative stress produced by NOX pathways and mitochondria directly contribute to a sarcolemmal loss of nNOS during mechanical unloading, enhancing proteolysis, and suppressing protein synthesis. The rodent hindlimb unloading model and novel fractional synthesis and degradation will be used to test our hypotheses. Specific Aim 1 will Identify NOX-dependent signaling on nNOS translocation and muscle fiber atrophy with hindlimb unloading. Specific Aim 2 will determine directly the role of mitochondrial and cytosolic oxidative stress in movement of nNOS away from the muscle cell membrane and disruption of DGC with mechanical unloading.<br>In short, our primary objective is to identify the trigger by which skeletal muscles switch from being an anabolic to a catabolic machine during spaceflight, causing muscle atrophy and weakness. We propose that oxidative stress produced by a NOX pathway and mitochondria move the master switch protein (nNOS) away from the cell membrane, which slows protein synthesis and accelerates breakdown of protein. Our study would directly address high-priority gaps in "Recapturing a Future for Space Exploration" including the role of reactive oxygen species in protein balance and maintaining the slow contractile phenotype. Our grant proposal would also forward our understanding of (1) alterations in the subsarcolemmal cytoskeleton and membrane domains, (2) sensing and transduction of loading and gravity, (3) protein turnover, and (4) regulatory mechanisms that govern alterations in skeletal muscle, all programmatic emphases and goals of the current Space Biology NRA (NASA Research Announcement). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Bedrest, casting, and spaceflight cause significant skeletal muscle atrophy and weakness as a result of disuse. Prolonged spaceflight presents great challenges to the physical performance and health of astronauts as a result of microgravity, or reduced gravitational forces (i.e., loads). Mechanical unloading and extreme disuse during spaceflight produce rapid and dramatic skeletal muscle and weakness. Prolonged disuse also predisposes skeletal muscle to damage when landing on Mars, in an EVA (extra-vehicular activity), or returning to Earth. Furthermore, chronic disuse also contributes to and exacerbates underlying chronic pathology, including insulin resistance and cardiovascular disease. As a result crew health, safety, and completion of mission objectives are placed in jeopardy. While the pathways that cause atrophy and weakness due to disuse remain uncertain, proteins that sense changes in forces that skeletal muscles must produce may be the key. One of these proteins is nitric oxide synthase (nNOS), which is positioned near the muscle cell membrane. During spaceflight as well as aging and some types of muscular dystrophy, nNOS appears to be transported away from the cell membrane where it transforms from a muscle building to a muscle wasting agent. We will focus on oxidative stress, produced by excess free radicals produced by proteins near the cell membrane (ex. NOX) and mitochondria and how they affect nNOS transport as well as skeletal muscle atrophy.<br>Understanding how these key membrane proteins are regulated will not only aid us in our treatment of disuse-related atrophy, but also will allow development of new therapeutics combating sedentary lifestyle and possible application to aging, Type 2 diabetes, and muscular dystrophies. Indeed, changes that are altered during spaceflight and ground models of spaceflight also appear to change with aging and diseases that cause skeletal muscle wasting and weakness. For example, loss of nNOSμ from the sarcolemma is a commonality abound spaceflight/disuse, aging, Type 2 diabetes, and Duchenne muscular dystrophy. Oxidative stress appears to be at the root of nNOSμ loss in theses muscle myopathies. Out laboratory seeks to understand the underlying redox mechanisms that leader to perturbation in nNOS function and thus myopathy. |
| | A. The Role of Oxidative Stress in Akt-mTOR Anabolic Signaling in Unloaded Skeletal Muscle<br>Skeletal muscles are dynamic mechanical and metabolic machines that adapt quickly to changes in their mechanical environment by altering rates in protein synthesis and protein degradation, and thus mass. Removal of mechanical loading during the microgravity (μG) of spaceflight or bedrest causes rapid atrophy of affected skeletal muscles. Recently, we identified that translocation of the mu-splice variant of neuronal nitric oxide synthase (nNOSμ) and downstream activation of FoxO3a were redox-sensitive phenomena (Lawler et al., AJP 2014 Apr 1;306(7):R470-82). Here, we tested the hypothesis that reactive oxygen species (ROS) contributed to perturbations in Akt-mTOR signaling using the rodent hindlimb unloading (HU) model of microgravity. The superoxide dismutase/catalase mimetic EUK-134 (10 mg/kg/day) was used as an antioxidant intervention in a 7-day protocol (n=6/group) using Fischer-344 (F344) rats: control ambulatory (CON), 7 days of hindlimb unloading (HU), and HU + EUK-134 (HUEUK). We found that EUK-134 reduced oxidative stress, dephosphorylation of FoxO3a, and mitigated muscle fiber atrophy and fiber-type shift from slow to fast-twitch in the rat soleus muscle. Furthermore, Akt phosphorylation (Ser473)/total Akt ratio and phosphorylated mTOR (Ser2448)/total mTOR ratio were greater in the HUEUK group than in HU solei. However, |

EUK-134 did not rescue p70S6K phosphorylation (Thr389) during HU. Membrane GLUT-4+ staining increased with HU but was abrogated by EUK-134, consistent with a fiber-type shift during HU, and inconsistent with increased insulin resistance by mechanical unloading. Therefore, we conclude that oxidative stress alters both anabolic (Akt-mTOR) and catabolic (FoxO3a) signaling in the unloaded soleus muscle, during a ground μG model.

B. Effect of EUK-134 on nNOS translocation and membrane repair proteins regulation in the rat soleus during 7 days of mechanical unloading

Skeletal muscles exhibit high adaptability in response to mechanical stimuli. Decreases in mechanical loading, such as bedrest or spaceflight, alter the balance between protein synthesis and degradation, resulting in muscle atrophy. Recent studies revealed that oxidative stress contributes to unloading-induced muscle atrophy via translocation of the mu-splice variant of neuronal nitric oxide synthase (nNOSμ), calpains, and activation of FoxO3a. While the transport system responsible for nNOSμ translocation is unclear, nNOSμ appears to detach during unloading from a-syntrophin and dystrophin, but remains tethered to dysferlin. However, the role of membrane repair proteins in unloading-induced nNOS translocation is unknown. We hypothesized that ROS would affect membrane repair proteins associated with nNOSμ translocation in unloaded skeletal muscle. The superoxide dismutase/catalase mimetic EUK-134 (10 mg/kg/day) was used as an antioxidant intervention in a 7-day protocol (n=6/group) using Fischer-344 (F344) rats: control ambulatory (CON), 7 days of hindlimb unloading (HU), and HU + EUK-134 (HUEUK). We used cytosolic and crude membrane protein expression of nNOS to observe translocation of nNOS. We also determined abundance and location of dysferlin and caveolin-3, a caveolar protein that can inhibit nNOS. We found that EUK-134 provided significant protection against loss of nNOS from the crude membrane fraction, with levels greater than the HU group but similar with controls. Conversely, HU caused a significant increase in cytosolic nNOS abundance, while cytosolic nNOS abundance in the HUEUK group was not significantly greater than CON. Caveolin-3 and dysferlin protein levels were higher in the unloaded soleus in both the membrane and cytosolic fractions, while caveolin-3 upregulation was mitigated by EUK-134 intervention. Our results suggest that oxidative stress plays a role in nNOS translocation, possibly guided by dysferlin and caveolin-3.

C. Nox2 Inhibition Prevents Skeletal Muscle Atrophy and nNOS Translocation in Hindlimb Unloaded Rats.

During mechanical unloading (e.g., spaceflight, bedrest), skeletal muscle undergoes an atrophic process, rapidly losing mass and cross-sectional area, with a concomitant decrease in force generating capacity and shift from type I (slow twitch) to type II (fast twitch) muscle fibers. Our research focuses on the generation of reactive oxygen species (ROS) during muscle atrophy, and how inhibition of ROS production may help ameliorate the effects of unloading and atrophy on skeletal muscle. We have identified translocation of the mu splice variant of neuronal nitric oxide synthase (nNOSμ) into the cell as a signaling event in atrophy. In addition, upregulation of the membrane bound protein NADPH oxidase 2 (NOX2) could be a key source of cellular ROS production during mechanical unloading. We hypothesized that use of the selective NOX2 inhibitor gp91ds-tat would prevent the deleterious effects of unloading on skeletal muscle.

**Task Progress:**

We utilized the rodent hindlimb unloading model as a ground analog for microgravity. Fifteen Fischer-344 rats were divided into three groups (n=5): ambulatory controls (CON), hindlimb unloaded animals (HU), and hindlimb unloaded animals treated with gp91ds-tat (HU-G).

Treatment with gp91ds-tat had effects on markers of muscle morphology and nNOSμ translocation. Using one-way ANOVA with SNK post hoc test, we determined that gp91ds-tat offered significant protection against atrophy when comparing intervention animals to their unloaded counterparts ($p < 0.05$), and approached significance. Further, gp91ds-tat prevented nNOSμ translocation, as demonstrated using confocal micrographs, while also ameliorating the nNOS activity inside the muscle cell, as measured using NADPH diaphorase staining ($p < 0.05$). Based on these results, we conclude that NOX2 plays a role in nNOSμ translocation away from the sarcolemma and muscle atrophy, and that inhibition of this enzyme could be an avenue by which to prevent the harmful effects of unloading on skeletal muscle.

D. Effect of Fish Oil and Curcumin Supplementation on Muscle Cross Sectional Area, Anabolic Signaling.

Skeletal muscle is a highly adaptable tissue capable of remodeling when dynamic stress is altered such as mechanical loading/unloading and stretch. When muscle is subjected to an unloaded state associated with bedrest, immobilization, or microgravity the muscle cross sectional area (CSA), and fiber-type shift from slow to fast, and impairs force production. In addition, muscle fiber-type shifts from slow to fast-twitch fibers. Unloading also results in a downregulation of anabolic signaling and heat shock proteins which further exacerbate these morphological changes. Our lab recently showed reactive oxygen species (ROS) are causal in translocation of neuronal nitric oxide synthase (nNOSμ) away from the cell membrane, muscle fiber atrophy, and fiber-type shift. Nutritional supplements fish oil and curcumin improves anabolic signaling, glutathione, and heat shock proteins, respectively. We hypothesized that fish oil, rich in omega-3-fatty acids, combined with polyphenol curcumin protect anabolic (Akt pathway) signaling and heat shock proteins in the rat soleus muscle, concomitant with protection of morphology, as a synergistic countermeasure. For this study C57BL/6 mice were divided into 3 groups (n=6/group): ambulatory controls w/3% corn oil in diet (CON), hindlimb unloading w/3% corn oil in diet (HU), and hindlimb unloading w/3% fish oil and 1% curcumin in diet. Fish Oil and Curcumin (FOC) treatments began 7 days prior to HU and mice were anesthetized following 10 days of HU. FOC mitigated the unloading-induced decrease in CSA and prevented the fiber-type shift normally. FOC also rescued anabolic signaling (Akt phosphorylation, p70S6K phosphorylation) and increased the abundance of HSP70. Therefore, we concluded that FOC prevents muscle atrophy in concert with the ability to boost heat shock proteins and anabolic signaling in an unloaded state.

E. AT1 receptor blockade attenuates hindlimb unloading-induced modifications of Nox2, nNOS, and FoxO3a signaling in skeletal muscle.

Extended duration of reduced mechanical loading results in weakening and atrophy of skeletal muscles. Disuse-induced atrophy is due in part to increased production of reactive oxygen species (ROS). For instance, ROS are causal in key disuse atrophy signaling events such as sarcolemma dislocation of nNOS and activation of FoxO3a. Upregulation of the Nox2 isoform of NADPH oxidase has been implicated in the disuse-associated increase in ROS production. Evidence indicates that activation of angiotensin II type 1 receptor (AT1R) is an upstream event leading to Nox2 complex formation and activity. Here, we used an AT1R blocker (losartan) to test the hypothesis that AT1R activation is an upstream signaling event driving Nox2 complex activity, nNOS dislocation, FoxO3a activation, and thus the slow-to-fast fiber type shift and atrophy in the unloaded soleus. Rats were divided into 4 groups: ambulatory control (CON), ambulatory + Losartan (40 mg/kg/day) (CONL), 7-days hindlimb unloaded (HU), and HU + Losartan (HUL). We report that administration of losartan attenuated unloading-induced phenotypic alterations in the soleus muscle, including the

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| | shift in fiber-type and reductions in muscle fiber cross-sectional area (CSA). Losartan administration ameliorated increased ROS levels and the upregulation and activity of Nox2 following unloading. Furthermore, AT1R blockade during unloading mitigated nNOS dislocation away from the sarcolemma. In addition, losartan treatment during disuse offered significant protection against the increase in nuclear localized FoxO3a. We conclude that AT1R blockade attenuates disuse atrophy by inhibiting Nox2 signaling, thereby lessening the extent of nNOS dislocation related signaling. |
| **Bibliography Type:** | Description: (Last Updated: 11/13/2025) |
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